

Class 1: Systems of Linear Equations

A linear equation is any equation of the form

Examples:

What does the graph of a linear equation look like?

What does it mean to solve a linear equation? What does it mean to solve a system of linear equations?

What are the three types of solutions for a system of linear equations?

The basic strategy to solve a system of linear equations is

Three operations that can simplify a system without changing the solution set:

1. Replace
2. Swap
3. Scale

Example: Simplify the system and find the solution set.

$$\begin{array}{rcl} 2x + 3y - 2z & = & -10 \\ x + 2y & = & -3 \\ -x & + & 6z = 9 \end{array}$$

Linear Systems and Matrices

Equations

$$x - 2y - z = 4$$

$$2x + 4y - 5z = 6$$

Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 4 \\ 2 & 4 & -5 & 6 \end{array} \right]$$

Operations on equations in the system correspond to operations on the rows of the matrix. The strategy to solve a system is to use row operations to convert the original augmented matrix into **row echelon form** or **reduced row echelon form**.

Row echelon form

1. Any row of all zeros must be below any row with a nonzero entry.
2. The leading entry of a row must be further left than a leading entry of any row below it.
3. All entries in a column below the leading entry must be zero.

Reduced row echelon form

4. Each leading entry must be a 1.
5. All entries above and below a leading 1 must be zero.

Practice For each matrix decide if it is in echelon or reduced echelon form or neither.

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

$$E = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$F = \begin{bmatrix} 1 & 2 & 0 & 4 & 0 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$G = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 0 & 3 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$H = \begin{bmatrix} 0 & 2 & -6 \\ 0 & -1 & 1 \end{bmatrix}$$

$$M = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

